

Physics of Semiconductor Devices

ECE210: PSD



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week 05



Reciprocal lattice



the reciprocal lattice represents the Fourier transform of another lattice (i.e. a Bravais lattice in real space).

Other names: Fourier space, k-space, or momentum space

- The electronic number density $n(\mathbf{r})$ in the crystal is a periodic function in space:

$$n(\mathbf{r}) = n(\mathbf{r} + \mathbf{R})$$

with period $\mathbf{R}$ equal to a

Direct Lattice Translation Vector:

$$\mathbf{R} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$$

- $n(\mathbf{r})$ can be expressed as a Fourier series expansion and In **3 Dimensions**, **the Fourier coefficient of the number density has the form**

$$n(\mathbf{r}) = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{K} \cdot \mathbf{r}} \rightarrow n_{\mathbf{G}} = \frac{1}{V_c} \int_0^a dV n(\mathbf{r}) e^{-i\mathbf{K} \cdot \mathbf{r}}$$

The vectors $\mathbf{K}$ are called
Reciprocal Lattice Vectors

Reciprocal lattice



- As we said, the electronic density $\mathbf{n}(\mathbf{r})$ is required to be invariant (periodic) under lattice translations:

$$\mathbf{n}(\mathbf{r}) = \mathbf{n}(\mathbf{r} + \mathbf{R}) \quad (1)$$

- That is, it must satisfy:

$$\begin{aligned} n(\mathbf{r}) &= n(\mathbf{r} + \mathbf{R}) \\ \Rightarrow \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{K} \cdot \mathbf{r}} &= \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{K} \cdot (\mathbf{r} + \mathbf{R})} \\ \Rightarrow \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{K} \cdot \mathbf{r}} &= \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{K} \cdot \mathbf{r}} \cdot \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{K} \cdot \mathbf{R}} \\ \Rightarrow e^{i\mathbf{K} \cdot \mathbf{R}} &= 1 \quad (2) \end{aligned}$$

- Only The Set of Reciprocal Lattice Vectors $\mathbf{K}$ that satisfy both (1) & (2) lead to an electronic number density $\mathbf{n}(\mathbf{r})$ that is invariant under lattice translations.

Reciprocal lattice



Direct Lattice Vector:

$$\mathbf{R} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$$

Reciprocal Lattice Vector:

$$\mathbf{K} = m\mathbf{a}^* + n\mathbf{b}^* + o\mathbf{c}^*$$

$$a^* = 2\pi \frac{b \times c}{a \cdot (b \times c)}$$

$$b^* = 2\pi \frac{c \times a}{a \cdot (c \times b)}$$

$$c^* = 2\pi \frac{a \times b}{c \cdot (a \times b)}$$

Reciprocal lattice is the Fourier transform of the real lattice

**We will construct reciprocal lattices
in 1D**

Real Lattice



Reciprocal Lattice



The Set of Reciprocal Lattice Vectors $\mathbf{K}$ is also Bravais Lattice

Significance of Reciprocal Lattice Concept

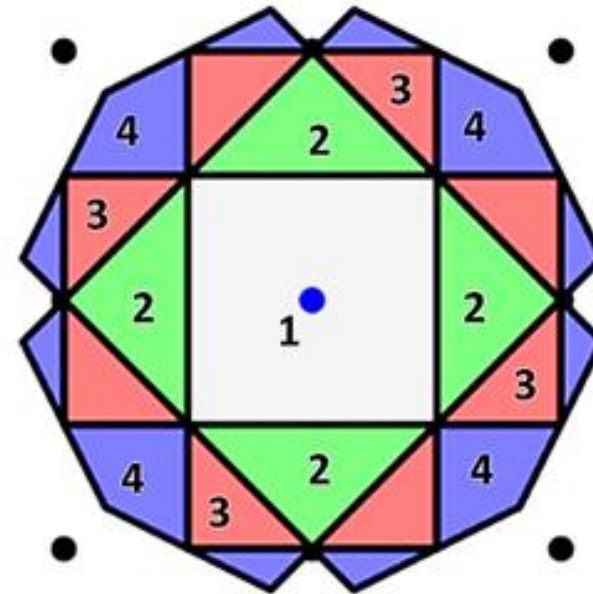
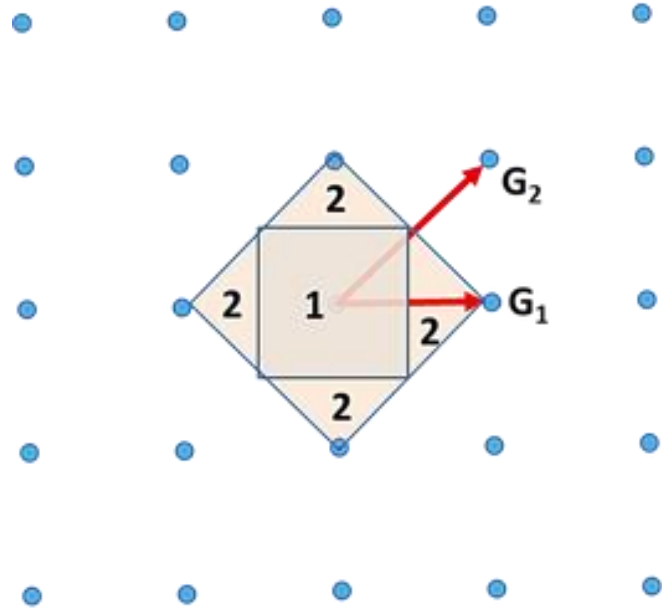


- ❑ An X-Ray diffraction pattern of the lattice Can be interpreted as a map of the reciprocal lattice of the crystal
- ❑ Momentum is directly proportional to wavevector and is inversely proportional to the wavelength. Therefore **each point of the reciprocal lattice represents the direction and the magnitude of momentum**, which is extremely convenient for solving Schrodinger Equation.
- ❑ From reciprocal lattice, we can conceive the concept of **Brillouin Zone**, which is very **helpful to explain several properties (band structure, effective mass, mobility etc.)** of semiconductor devices.

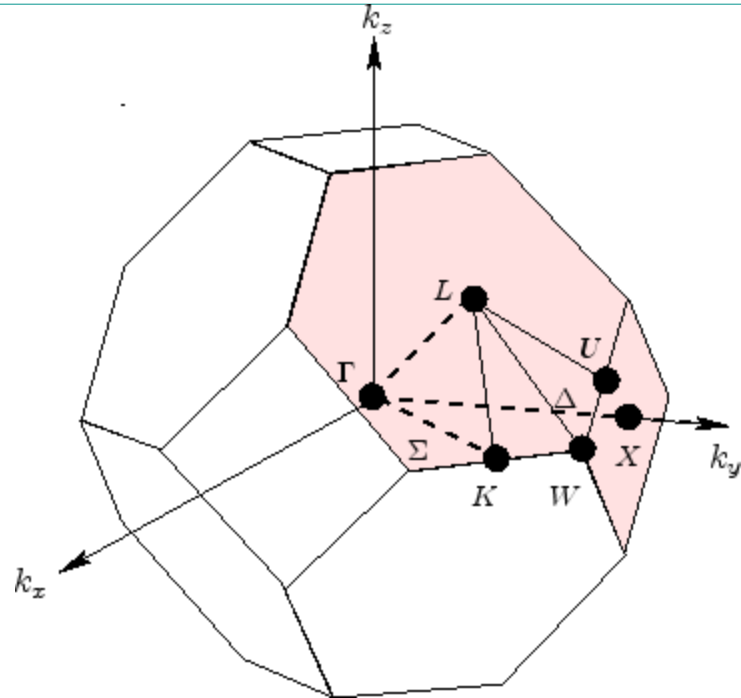
Brillouin zone



The Wigner-Seitz cell of the reciprocal lattice is conventionally referred as the Brillouin zone.



High Symmetry points in the Brillouin zone of a fcc lattice



$$\Gamma = (0, 0, 0)$$

$$X = (1, 0, 0) \equiv (2\pi/a, 0, 0)$$

$$L = (1/2, 1/2, 1/2) \equiv (\pi/a, \pi/a, \pi/a)$$

Significance of Brillouin Zone

- ❑ A Brillouin zone is a property of a crystal. This geometrical shape can be considered to contain the valence electrons of the crystal. Its planes define, in momentum space (k-space), the location of the band gap.
- ❑ Points of symmetry on the Brillouin zone are given particular importance especially when determining the bandstructure of the material.

Study at home:

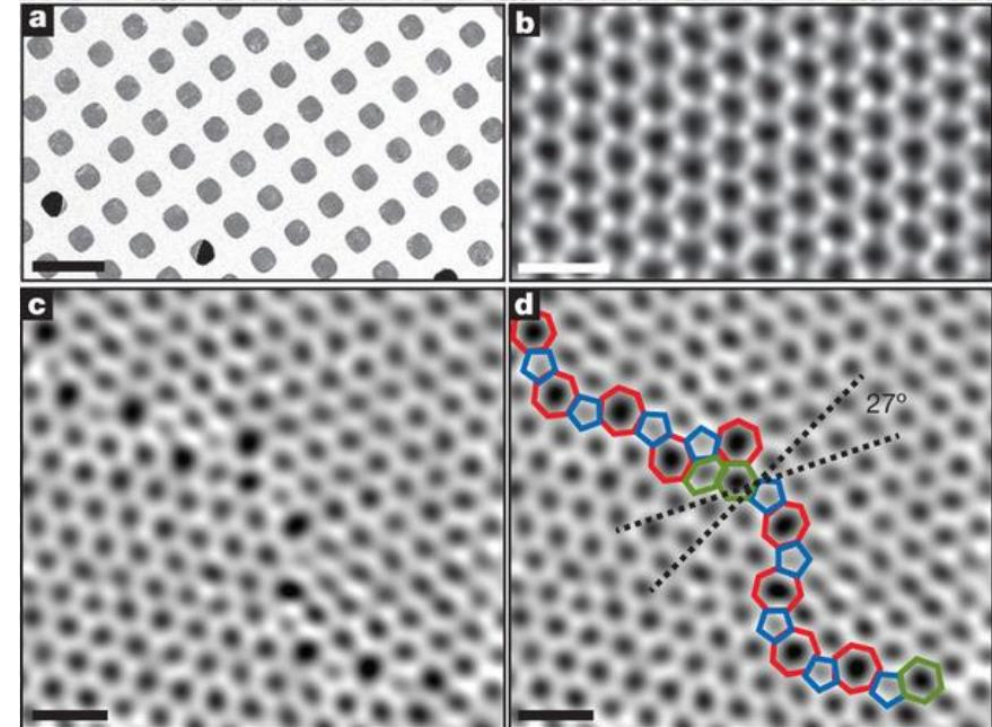
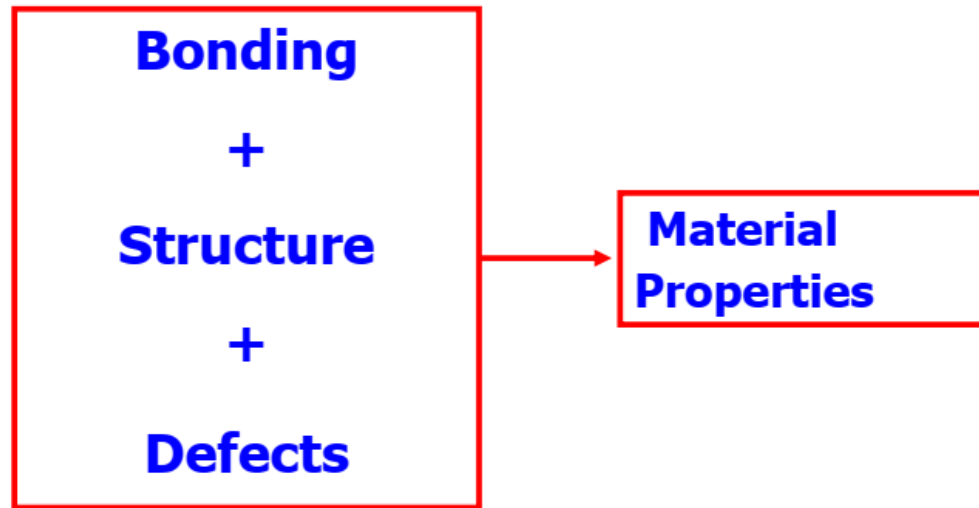
- ☐ Miller Indices
- ☐ Bragg's Law
- ☐ Ewald Experiment
- ☐ Laue Method



Defects in Crystalline Solids



Defects have a profound impact on the macroscopic properties of materials

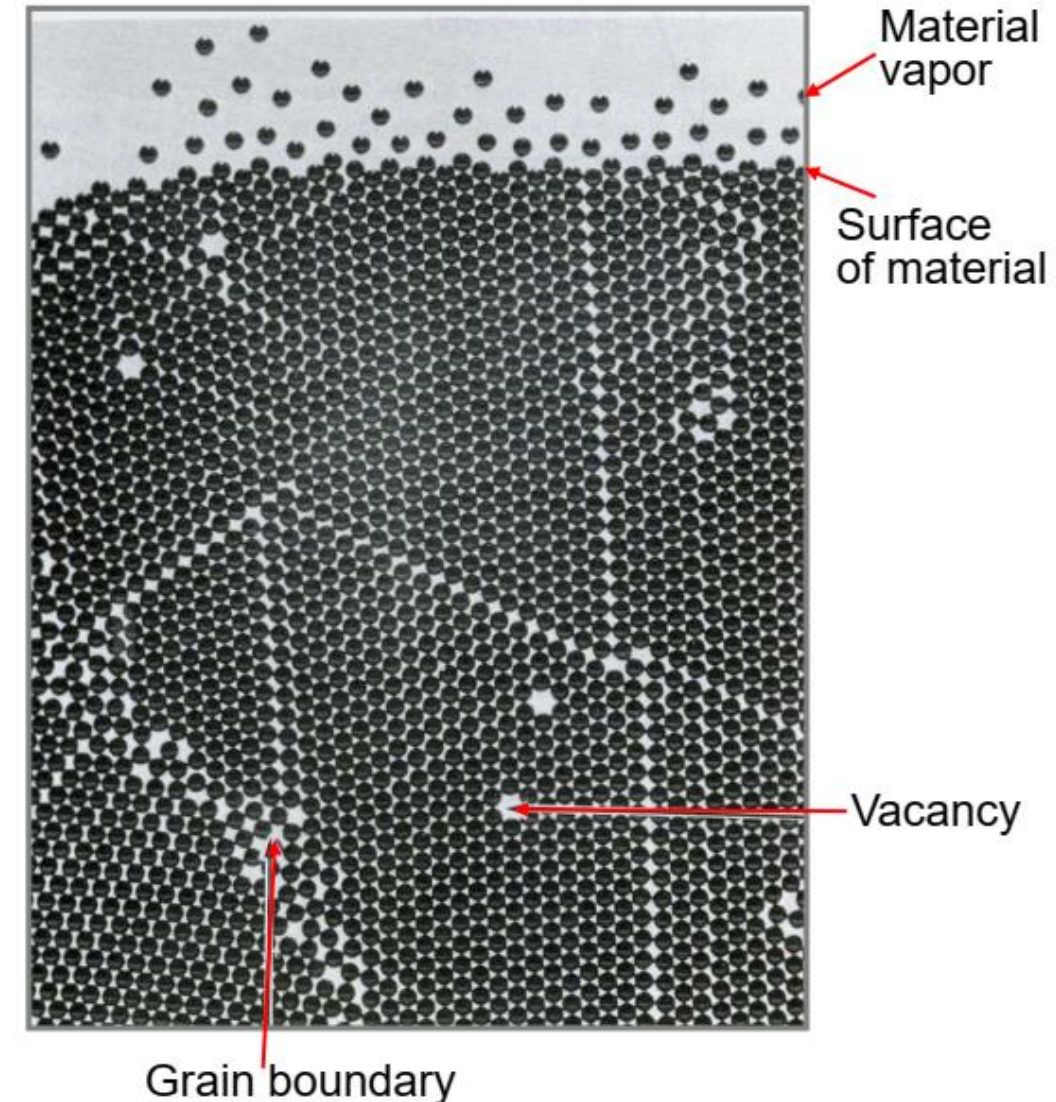


- **Real crystals** are never perfect: there are **always defects**!
- **Controlling** the defects is one of **the main** goal of materials science and engineering!

ISSUES TO ADDRESS...

- What **types of defects** arise in solids?
- Can the number and type of defects be varied and **controlled**?
- How do defects **affect material properties**?
- Are all defects **undesirable**?

- An ideal crystal is regarded as a perfect crystal which has the same unit cell and contains the same Lattice Point (LP) throughout the crystals.
- Crystals always have a regular periodic arrangement of atoms. Any deviation from this periodicity is known as defects or imperfections in crystals.
- Definition: The deviation from the perfect periodicity of atomic arrays in crystals is known as crystal defects.
- Any deviation from this makes it imperfect (defects)
- Classification of crystal Varies from one author to another



Types of Imperfections



■ Defects are classification according to geometry (point, line or plane).

■ ∃ 4 categories

- Point defects - 0D
- Line defects - 1D
- Area defects - 2D
- Volume defects - 3D

Point defects

- Vacancy **atoms**
- Self-Interstitial **atoms**
- Impurities: Substitutional and Interstitial **atoms**

Line defects

- Dislocations

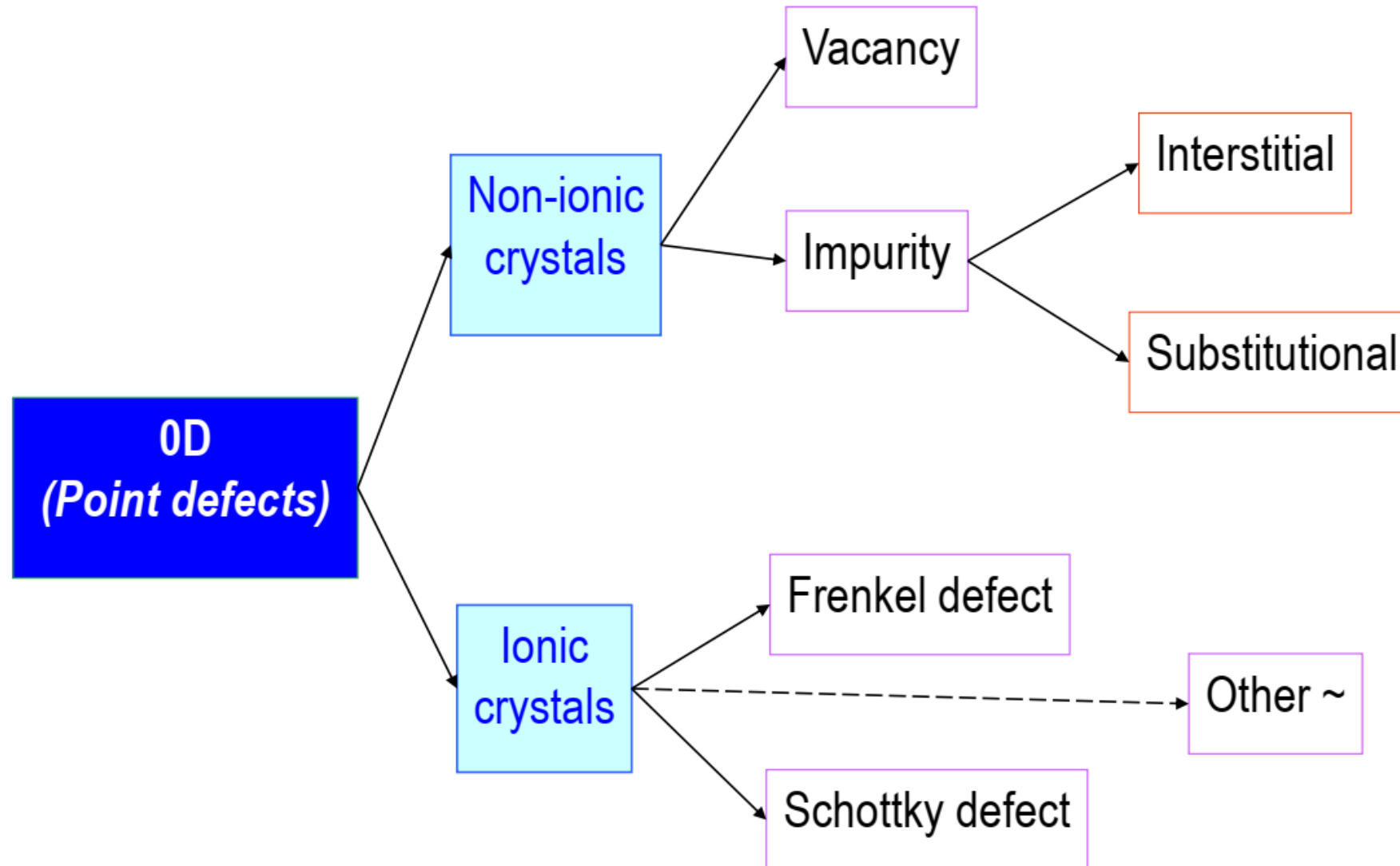
Area defects

- Grain Boundaries

Volume defects

Are distributed within volume of material
(pores, cracks, foreign inclusions etc)

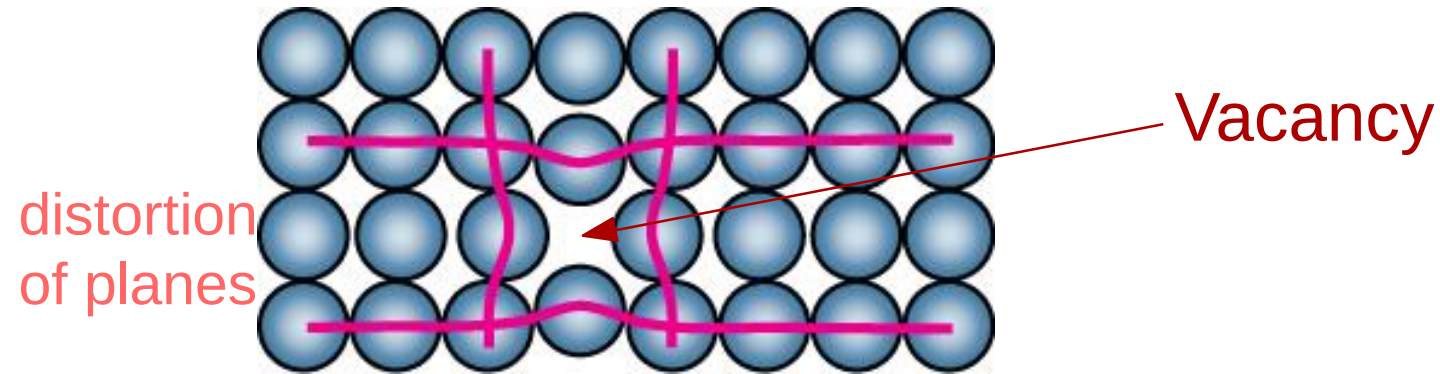
Point defects



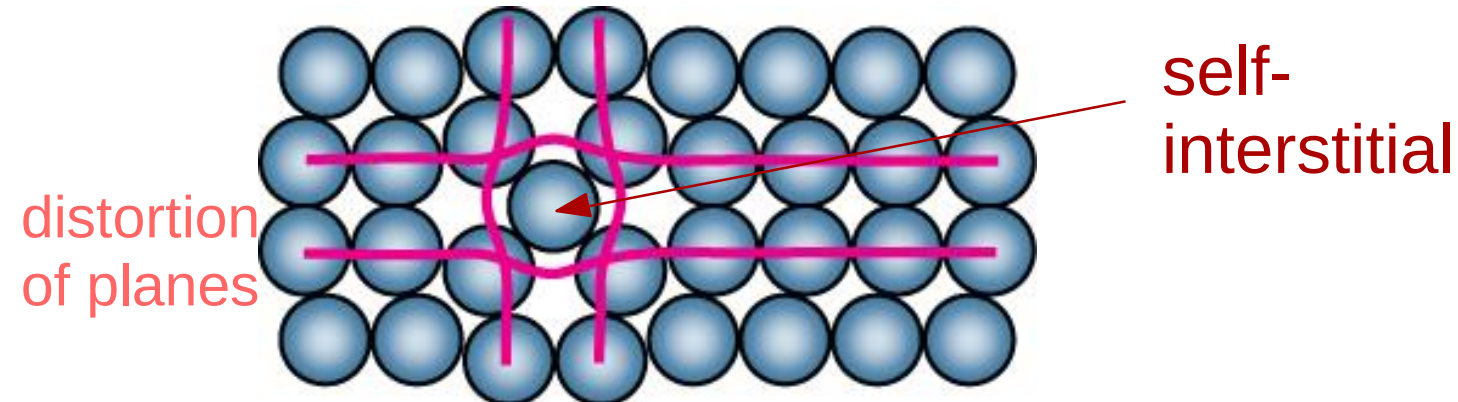
Point Defects in Metals



- **Vacancies:**
-vacant atomic sites in a structure.



- **Self-Interstitials:**
-"extra" atoms positioned between atomic sites.

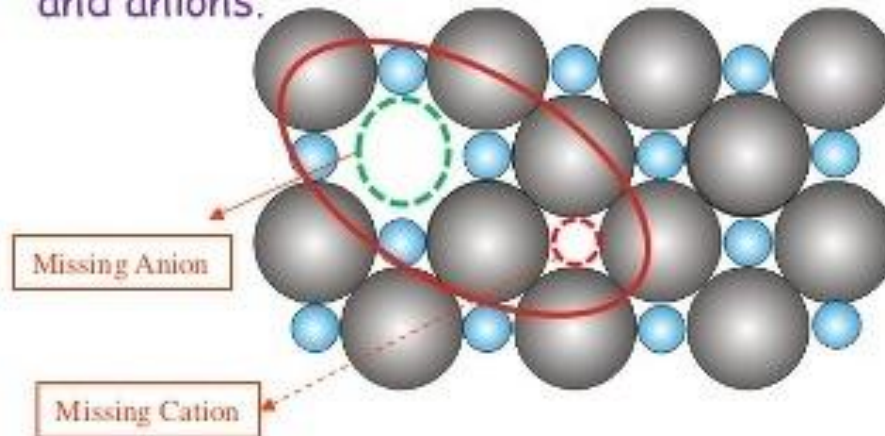


Schottky and Frenkel defect



Schottky Defect:

- ❖ Forms when oppositely charged ions leave their lattice sites, creating vacancies.
- ❖ These vacancies are formed in stoichiometric units, to maintain an overall neutral charge in the ionic solid.
- ❖ Density of the solid crystal is less than normal
- ❖ Occurs only when there is small difference in size between cations and anions.



Frenkel Defect:

- ❖ Smaller ion (usually the cation) is displaced from its lattice position to an interstitial site.
- ❖ Creates a **vacancy defect** at its original site and an **interstitial defect** at its new location.
- ❖ Does not change the density of the solid.
- ❖ Shown in ionic solids with large size difference between the anion and cation.

